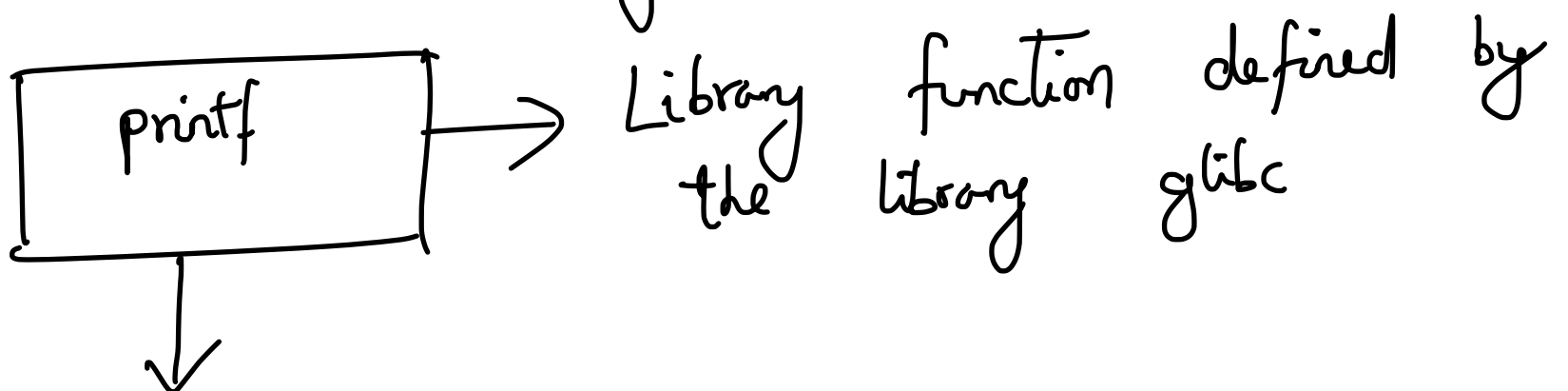
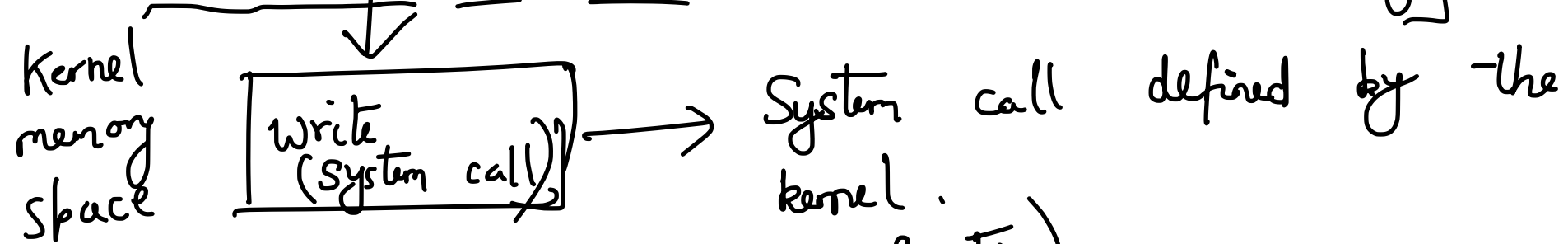
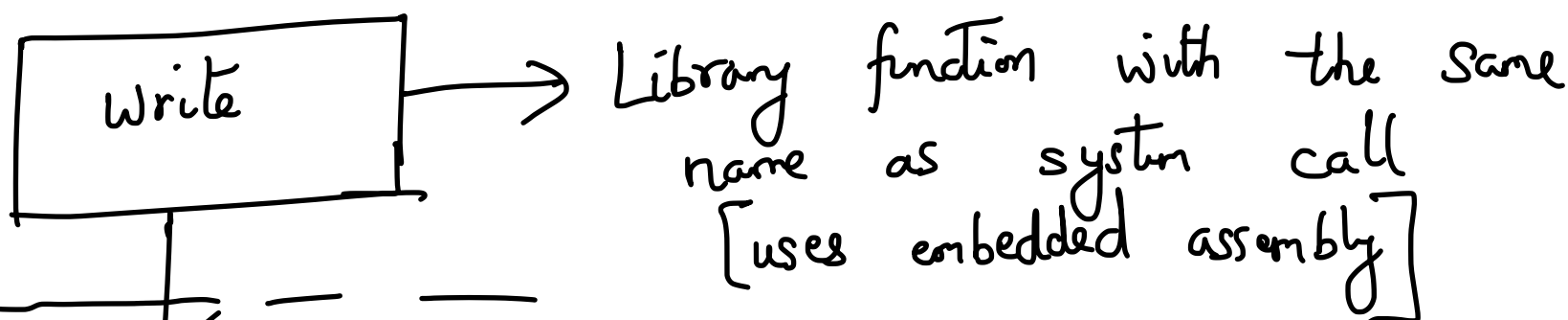


- 1) Assembly
 - 2) Process Management
 - Process Data Structures
 - Process System ~~Call~~ Calls
 - Threads
 - Process Scheduling
 - 3) Structure of Kernel → Layered kernels, microkernel
 - 4) How do system calls work?
 - 5) How does the bootup process work?
-

System calls

→ A mechanism of a user program communicating with the OS.





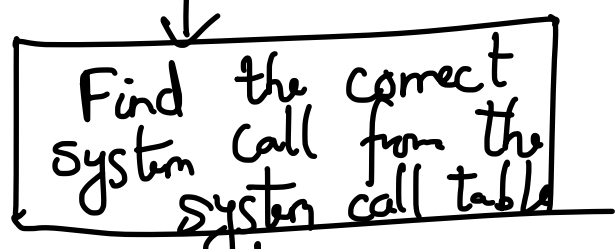
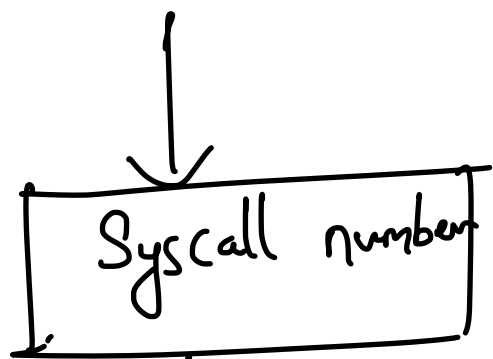
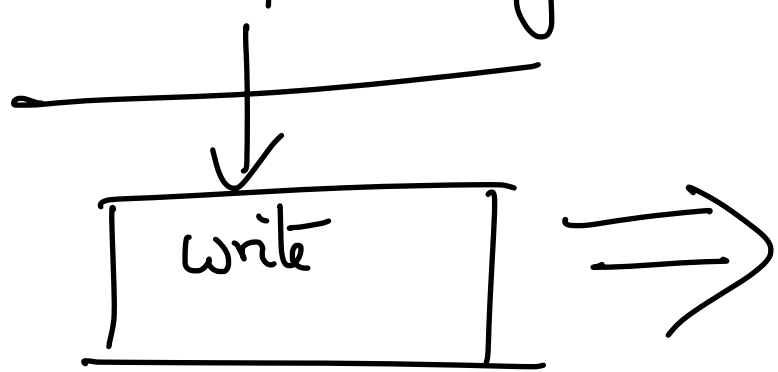
(no access to C library functions)

copy-to-user to copy the data to

the user's memory space from kernel's memory space.

copy-from-user to copy the data ~~from~~ to the kernel's memory space from user's memory space.

Defined by the Linux kernel itself



This is done in the process context, so multiple processes can make similar



requests at the same time.

Check the arguments
whether they are valid



copy-from-user



processing of requests
by making requests
to device drivers or
kernel data structures



copy-to-user



Copy return value
to rax register



Return control
to user program

Boot Loader

1) Processors start in 16-bit mode (real mode)

→ There is no proper memory management

→ BIOS (Basic I/O System)

→ Checks the hardware present;
on error it stops the boot process.

processor, RAM, keyboard

BIOS stored on ROM; needs to load data from disk

Small disk drivers are present in BIOS to access the disk.

→ First sector of the disk needs to be in a specific format if it has the OS loaded into it.

Linux has a boot loader called GRUB, ^(part of the kernel)
→ Boot loader takes control from the BIOS.

→ Reads each of the disks to find out the OSes present

→ If Linux kernel is chosen, then it loads the kernel's image into the main memory.

→ Set up the data structures required to enable memory management unit and then set a flag in the control register cr0 to move to 32-bit mode (protected mode).

Linux kernel ~~have~~ has a different levels of service called "init levels".

All GUI + other processes allowed → init level 5

Everything allowed except GUI → init level 3

Everything except GUI and network $\rightarrow$ init 2
Single (-root) user $\rightarrow$ init level 1

The init process is the first process created; the rest of the processes are spawned by using `fork()` and `exec()` one-by-one.

The command "init 0" shut downs the machine.

Daemon process : $\rightarrow$ A type of process that runs in background and provides different user services (eg: $\rightarrow$ networkd, sshd). Bash shell is not a daemon process.